

Huang (Raven) Huang

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Research Interests

Robot learning from human videos; real2sim2real transfer and scalable robot data generation; dexterous manipulation; multimodal perception; robot world models and motion planning.

Education

University of California, Berkeley 2020.08–2025.05
Ph.D. in Electrical Engineering and Computer Science
Advisor: Ken Goldberg

The University of Texas at Austin 2018.08–2020.05
M.S. in Mechanical Engineering

Shanghai Jiao Tong University 2014.09–2018.05
B.S. in Naval Architecture and Ocean Engineering

Research Appointments

Stanford University Apr. 2025–Present
Postdoctoral Researcher

Advisors: Jiajun Wu and Fei-Fei Li. Research on learning from human videos, multimodal perception with touch, and BEHAVIOR benchmark development.

Meta FAIR Mar.–Dec. 2025
Postdoctoral Researcher

Research on latent action-conditioned world models for cross-embodiment robot learning.

NVIDIA Robotics Lab, Seattle Sep. 2023–Feb. 2024
Research Scientist Intern

Research on learning-based robot motion planning, including diffusion-generated trajectories for motion optimization.

Honors and Awards

ICRA 2025 Best Paper Award on Robot Learning, co-author of “Robo-DM: Data Management for Large Robot Datasets.”

Microsoft Research Fellowship, 2026.

Teaching Experience

University of California, Berkeley One semester
Teaching Assistant, 106A/206A

The University of Texas at Austin Two years
Teaching Assistant

Peer-Reviewed Publications

* denotes equal contribution.

Chengshu Li, Mengdi Xu, Arpit Bahety, Hang Yin, Yunfan Jiang, **Huang Huang**, Josiah Wong, Sujay Garlanka, Cem Gokmen, Ruohan Zhang, Weiyu Liu, Jiajun Wu, Roberto Martín-Martín, Li Fei-Fei, “Momagen: Generating demonstrations under soft and hard constraints for multi-step bimanual mobile manipulation”, *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026

Huang Huang, Sriram Yenamandra, Arjun Majumdar, Elie Aljalbout, Tushar Nagarajan, Tsung-Yen Yang, Akshara Rai, Michael Rabbat, Li Fei-Fei, Jiajun Wu, Tingfan Wu, Franziska Meier, “Cross-Embodiment Robot Foundation World Models with Latent Actions”, *International Conference on Machine Learning (ICML)*, 2026.

Justin Yu, Letian Fu, **Huang Huang**, Karim El-Refai, Rares Andrei Ambrus, Richard Cheng, Muhammad Zubair Irshad, Ken Goldberg, “Real2Render2Real: Scaling Robot Data Without Dynamics Simulation or Robot Hardware”, *Conference on Robot Learning (CoRL)*, 2025

Mingxuan Wu*, **Huang Huang***, Justin Kerr, Chung Min Kim, Anthony Zhang, Brent Yi, Angjoo Kanazawa, “Predict-Optimize-Distill: A Self-Improving Cycle for 4D Object Understanding”, *International Conference on Computer Vision (ICCV)*, 2025

Huang Huang*, Fangchen Liu*, Letian Fu*, Tingfan Wu, Mustafa Mukadam, Jitendra Malik, Ken Goldberg, Pieter Abbeel, “OTTER: A Vision-Language-Action Model with Text-Aware Visual Feature Extraction”, *Forty-Second International Conference on Machine Learning (ICML)*, 2025

Letian Fu*, **Huang Huang***, Gaurav Datta*, Lawrence Yunliang Chen, William Chung-Ho Panitch, Fangchen Liu, Hui Li, Ken Goldberg, “ICRT: In-Context Imitation Learning via Next-Token Prediction”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2025

Andrew Goldberg, Kavish Kondap, Tianshuang Qiu, Zehan Ma, Letian Fu, Justin Kerr, **Huang Huang**, Kaiyuan Chen, Kuan Fang, Ken Goldberg, “Blox-Net: Generative Design-for-Robot-Assembly Using VLM Supervision, Physics Simulation, and a Robot with Reset”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2025

Kaiyuan Chen, Letian Fu, David Huang, Yanxiang Zhang, Lawrence Yunliang Chen, **Huang Huang**, Kush Hari, Ashwin Balakrishna, Ted Xiao, Pannag R. Sanketi, John Kubiawicz, Ken Goldberg, “Robo-DM: Data Management for Large Robot Datasets”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2025.

Huang Huang, Balakumar Sundaralingam, Arsalan Mousavian, Adithyavairavan Murali, Ken Goldberg, Dieter Fox, “DiffusionSeeder: Seeding Motion Optimization with Diffusion for Rapid Motion Planning”, *8th Annual Conference on Robot Learning (CoRL)*, 2024

Huang Huang, Satvik Sharma, Antonio Loquercio, Anastasios Angelopoulos, Ken Goldberg, Jitendra Malik, “Conformal policy learning for sensorimotor control under distribution shifts”, *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024

Letian Fu, Gaurav Datta*, **Huang Huang***, William Panitch*, Jaimyn Drake*, Joseph Ortiz, Mustafa Mukadam, Mike Lambeta, Roberto Calandra, Ken Goldberg, “A touch, vision, and language dataset for multimodal alignment”, *2024 International Conference on Machine Learning*, 2024 (**Oral Presentation**)

Adam Rashid, Chung Kim, Justin Kerr, Letian Fu, Kush Hari, Ayah Ahmad, Kaiyuan Chen, **Huang Huang**, Marcus Gualtieri, Michael Wang, Christian Juetten, Nan Tian, Liu Ren, Ken Goldberg, “Lifelong LERF: Local 3D Semantic Inventory Monitoring Using FogROS2”, *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024

Qinxi Yu, Masoud Moghani, Karthik Dharmarajan, Vincent Schorp, William Panitch, Jingzhou Liu, Kush Hari, **Huang Huang**, Mayank Mittal, Ken Goldberg, Animesh Garg, “ORBIT-Surgical: An Open-Simulation Framework for Learning Surgical Augmented Dexterity”, *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024

Huang Huang, Antonio Loquercio, Ashish Kumar, Neerja Thakkar, Ken Goldberg, Jitendra Malik, “Manipulator as a tail: Promoting dynamic stability for legged locomotion”, *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024

Lawrence Yunliang Chen, **Huang Huang**, Ken Goldberg, “Optimal Arrangement and Rearrangement of Objects on Shelves to Minimize Robot Retrieval Cost”, *IEEE Transactions on Automation Science and Engineering*, 21(3):2184–2198, 2024.

Karthik Dharmarajan, Will Panitch, Muyan Jiang, Kishore Srinivas, Baiyu Shi, Yahav Avigal, **Huang Huang**, Thomas Low, Danyal Fer, Ken Goldberg, “Automating vascular shunt insertion with the dvrk surgical robot”, *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023

Karthik Dharmarajan, Will Panitch, Baiyu Shi, **Huang Huang**, Lawrence Chen, Thomas Low, Danyal Fer, Ken Goldberg, “A trimodal framework for robot-assisted vascular shunt insertion when a supervising surgeon is local, remote, or unavailable”, *2023 International Symposium on Medical Robotics (ISMR)*, 2023

Letian Fu, **Huang Huang**, Lars Berscheid, Hui Li, Ken Goldberg, Sachin Chitta, “Safe self-supervised learning in real of visuo-tactile feedback policies for industrial insertion”, *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023

Satvik Sharma*, **Huang Huang***, Kaushik Shivakumar, Lawrence Chen, Ryan Hoque, Brian Ichter, Ken Goldberg, “Semantic mechanical search with large vision and language models”, *7th Annual Conference on Robot Learning*, 2023

Karthik Dharmarajan, Will Panitch, Baiyu Shi, **Huang Huang**, Lawrence Chen, Masoud Moghani, Qinxu Yu, Kush Hari, Thomas Low, Danyal Fer, Animesh Garg, Ken Goldberg, “Robot-Assisted Vascular Shunt Insertion with the dVRK Surgical Robot”, *Journal of Medical Robotics Research*, 2023

Justin Kerr*, **Huang Huang***, Albert Wilcox, Ryan Hoque, Jeffrey Ichnowski, Roberto Calandra, Ken Goldberg, “Learning Self-Supervised Representations from Vision and Touch for Active Sliding Perception of Deformable Surfaces”, *Robotics: Science and Systems XIX*, 2023

Huang Huang, Michael Danielczuk, Chung Kim, Letian Fu, Zachary Tam, Jeffrey Ichnowski, Anelia Angelova, Brian Ichter, Ken Goldberg, “Mechanical search on shelves using a novel “bluction” tool”, *2022 International Conference on Robotics and Automation (ICRA)*, 2022

Lawrence Chen, **Huang Huang**, Michael Danielczuk, Jeffrey Ichnowski, Ken Goldberg, “Optimal shelf arrangement to minimize robot retrieval time”, *2022 IEEE 18th International Conference on Automation Science and Engineering (CASE)*, 2022

Vincent Lim*, **Huang Huang***, Lawrence Chen, Jonathan Wang, Jeffrey Ichnowski, Daniel Seita, Michael Laskey, Ken Goldberg, “Real2sim2real: Self-supervised learning of physical single-step dynamic actions for planar robot casting”, *2022 International Conference on Robotics and Automation (ICRA)*, 2022

Justin Kerr, Letian Fu, **Huang Huang**, Yahav Avigal, Matthew Tancik, Jeffrey Ichnowski, Angjoo Kanazawa, Ken Goldberg, “Evo-nerf: Evolving nerf for sequential robot grasping of transparent objects”, *6th annual conference on robot learning*, 2022 (**Oral Presentation**)

Huang Huang, Letian Fu, Michael Danielczuk, Chung Kim, Zachary Tam, Jeffrey Ichnowski, Anelia Angelova, Brian Ichter, Ken Goldberg, “Mechanical search on shelves with efficient stacking and destacking of objects”, *The International Symposium of Robotics Research*, 2022

Lawrence Chen*, **Huang Huang***, Ellen Novoseller, Daniel Seita, Jeffrey Ichnowski, Michael Laskey, Richard Cheng, Thomas Kollar, Ken Goldberg, “Efficiently learning single-arm fling motions to smooth garments”, *The International Symposium of Robotics Research*, 2022

Brijen Thananjeyan, Justin Kerr, **Huang Huang**, Joseph Gonzalez, Ken Goldberg, “All you need is LUV: Unsupervised collection of labeled images using UV-fluorescent markings”, *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022

Yahav Avigal, Anna Deza, William Wong, Sebastian Oehme, Mark Presten, Mark Theis, Jackson Chui, Paul Shao, **Huang Huang**, Atsunobu Kotani, Satvik Sharma, Rishi Parikh, Michael Luo, Sandeep Mukherjee, Stefano Carpin, Joshua Viers, Stavros Vougioukas, Ken Goldberg, “Learning seed placements and automation policies for polyculture farming with companion plants”, *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021

Huang Huang, Marcus Dominguez-Kuhne, Vishal Satish, Michael Danielczuk, Kate Sanders, Jeffrey Ichnowski, Andrew Lee, Anelia Angelova, Vincent Vanhoucke, Ken Goldberg, “Mechanical search on shelves using lateral access x-ray”, *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021

Yahav Avigal, Vishal Satish, Zachary Tam, **Huang Huang**, Harry Zhang, Michael Danielczuk, Jeffrey Ichnowski, Ken Goldberg, “Avplug: Approach vector planning for unicontact grasping amid clutter”, *2021 IEEE 17th international conference on automation science and engineering (CASE)*, 2021

Harry Zhang, Jeffrey Ichnowski, Daniel Seita, Jonathan Wang, **Huang Huang**, Ken Goldberg, “Robots of the lost arc: Self-supervised learning to dynamically manipulate fixed-endpoint cables”, *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 2021

Bingham He, **Huang Huang**, Gray Thomas, Luis Sentis, “A complex stiffness human impedance model with customizable exoskeleton control”, *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2020

Huang Huang, Henry Cappel, Gray Thomas, Bingham He, Luis Sentis, “Adaptive compliance shaping with human impedance estimation”, *2020 American Control Conference (ACC)*, 2020

Bingham He, **Huang Huang**, Gray Thomas, Luis Sentis, “Complex stiffness model of physical human-robot interaction: Implications for control of performance augmentation exoskeletons”, *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019

Hang Xu, **Huang Huang**, Xiao-Hang Xu, Qiang Sun, “Modeling heat transfer of nanofluid flow in microchannels with electrokinetic and slippery effects using Buongiorno’s model”, *International Journal of Numerical Methods for Heat & Fluid Flow*, 2019

Hang Xu, **Huang Huang**, “Liquid flow in a porous channel with electrokinetic effects”, *Communications in Theoretical Physics*, 2018

Preprints

Yize Liu, **Huang Huang**, Yining Hong, Zijian Du, Zhi Cao, Li Fei-Fei, Jiajun Wu, “T²Mem: Learning Test-Time Memory for Robotics”, *arXiv:2609.36720*, 2026.

Youhui Wang, Yunzhu Li, Li Fei-Fei, Jiajun Wu[†], **Huang Huang**[†], “DexAgent: An Agentic Human2Sim2Robot Framework for Dexterous Manipulation with Self-Evolving Tool Library”, *arXiv:2609.35318*, 2026. ([†] *equal advising*)

Zhi Cao*, Howard Ji*, Kevin Zhang*, Kuangzhi Ge, Li Fei-Fei, Jiajun Wu[†], **Huang Huang**[†], “TrAct: Bridging Robot Control and Visual Prediction with Visual Tracks”, *arXiv:2608.24101*, 2026. ([†] *equal advising*)

Arthur Allshire, Himanshu Gaurav Singh, Ritvik Singh, Adam Rashid, Hongsuk Choi, David McAllister, Justin Yu, Yiyuan Chen, **Huang Huang**, Pieter Abbeel, Xi Chen, Rocky Duan, Phillip Isola, Jitendra Malik, Fred Shentu, Guanya Shi, Philipp Wu, Angjoo Kanazawa, “Scalable Behavior Cloning with Open Data, Training, and Evaluation”, *arXiv:2606.27375*, 2026.

Evans Han, Yunfan Jiang, Yingke Wang, Haoyue Xiao, **Huang Huang**, Jianwen Xie, Jiajun Wu, Li Fei-Fei, Ruohan Zhang, “StereoPolicy: Improving Robotic Manipulation Policies via Stereo Perception”, *arXiv:2605.09989*, 2026.

Letian Fu, Justin Yu, Karim El-Refai, Ethan Kou, Haoru Xue, **Huang Huang**, Wenli Xiao, Guanzhi Wang, Dantong Niu, Fei-Fei Li, Guanya Shi, Jiajun Wu, Shankar Sastry, Yuke Zhu, Ken Goldberg, Linxi “Jim” Fan, “CaP-X: A Framework for Benchmarking and Improving Coding Agents for Robot Manipulation”, *arXiv:2603.22435*, 2026.

Yining Hong, **Huang Huang**, Manling Li, Li Fei-Fei, Leonidas Guibas, Jiajun Wu, Yejin Choi, “Learning from Trials and Errors: Reflective Test-Time Planning for Embodied LLMs”, *arXiv:2602.21198*, 2026.

Vincent Lim, Ellen Novoseller, Jeffrey Ichnowski, **Huang Huang**, Ken Goldberg, “Policy-based bayesian experimental design for non-differentiable implicit models”, *arXiv preprint arXiv:2203.04272*, 2022

Professional Service

Associate Editor, IEEE International Conference on Robotics and Automation (ICRA), 2026.

Co-Organizer, [ICRA Workshop Pre-training for Robotics](#) , London, UK, 2023.

Co-Organizer, [ICRA Workshop Agile Robotics: From Perception to Dynamic Action](#) , Yokohama, Japan, 2024.

Session Chair, Surgical Robotics III at ICRA, Yokohama, Japan, 2024.